

FitBuddy – Detailed Project Planning

Project Planning and Execution Framework

This section defines how the FitBuddy project will be planned, organized, developed, tested, deployed, and maintained. FitBuddy is an AI-assisted fitness platform designed for athletes, coaches, and administrators, with features for personalized workout planning, workout tracking, progress monitoring, and feedback management.

1. Project Objectives

- Develop a user-friendly fitness platform for managing athlete profiles and workout activities.
- Provide structured and personalized workout plans based on goals and fitness information.
- Allow athletes to record workouts and monitor progress over time.
- Provide coaches and administrators with tools to manage athletes and workout plans.
- Use AI-assisted feedback to support useful, structured recommendations while keeping human review available.
- Build the system using modular components so future features can be added without major redesign.

2. Project Scope

In Scope	Out of Initial Scope
User registration, login, and role management	Clinical diagnosis or medical treatment
Athlete profile and fitness goals	Guaranteed health outcomes
Workout plan creation and assignment	Professional medical consultation
Exercise and workout-session tracking	Direct integration with every wearable device
Progress dashboards and basic analytics	Advanced predictive health models
Coach/admin management features	Large-scale social networking features
AI-assisted feedback and revision workflow	Fully autonomous fitness decision-making

3. Project Deliverables

Deliverable	Expected Output
Requirement Specification	Documented functional and non-functional requirements
UI/UX Design	Wireframes, screen layouts, navigation flow, and responsive design
Database Design	Entities, relationships, fields, constraints, and data-flow plan
Frontend	Responsive athlete, coach, and admin interfaces
Backend/API	Authentication, workout plans, sessions, progress, and feedback services
AI Feedback Module	Structured feedback generation and revision workflow
Testing Package	Test cases, defect records, and acceptance results
Deployment Package	Production-ready configuration and deployment documentation
Project Documentation	User guide, technical documentation, and final project report

4. Work Breakdown Structure (WBS)

Phase	Major Activities	Output
1. Initiation	Define problem, objectives, users, scope, constraints	Project charter
2. Requirements	Gather athlete/coach needs; define functional requirements	SRS / requirement list
3. Planning	Schedule tasks, assign responsibilities, estimate resources	Project plan
4. UI/UX	Create wireframes, navigation, responsive layouts	UI prototype
5. Architecture	Design frontend, backend, database, API, security	Architecture design
6. Development	Build modules and integrate services	Working application
7. AI/Feedback	Implement feedback workflow and validation	AI feedback module
8. Testing	Unit, integration, UI, security, performance, UAT	Test report
9. Deployment	Configure environment, database, hosting, monitoring	Deployed system
10. Documentation	Prepare technical and user documentation	Final documentation

5. Proposed Development Schedule

Week	Focus	Key Activities	Milestone
1	Initiation	Problem definition, objectives, scope, stakeholders	Project approved
2	Requirements	User stories, functional requirements, acceptance criteria	Requirements complete
3	UI/UX	Wireframes, navigation, dashboard and workout screens	Prototype ready
4	Architecture	Database, API, security, component structure	Architecture approved
5–6	Core Development	Authentication, profiles, roles, athlete dashboard	Core modules working
7–8	Workout System	Exercise catalogue, plan builder, assignment, sessions	Workout workflow complete
9	Progress	History, statistics, progress dashboard	Analytics module complete
10	AI Feedback	Feedback generation, review, revision history	AI workflow integrated
11	Testing	Functional, integration, security, UI and performance tests	Major defects resolved
12	Deployment	Production setup, documentation, final review	Release candidate

6. Team and Responsibility Plan

Role	Responsibilities
Project Manager	Scope, schedule, task tracking, meetings, risks, stakeholder communication
UI/UX Designer	User flows, wireframes, responsive layouts, usability
Frontend Developer	Pages, dashboards, forms, workout interfaces, API integration
Backend Developer	APIs, authentication, business logic, database integration
Database Developer	Schema, relationships, queries, indexes, backups
AI/ML Developer	Feedback workflow, prompt/model integration, output validation

QA Tester	Test planning, execution, defect tracking, regression testing
DevOps/Deployment	Environment configuration, deployment, monitoring, backups

7. Resource Planning

- **Human resources:** project manager, UI/UX designer, frontend developer, backend developer, database developer, QA tester, and AI/ML support.
- **Software resources:** code editor/IDE, version-control platform, database management tools, API testing tool, design/prototyping tool, and testing utilities.
- **Infrastructure:** development machines, application hosting, database hosting, domain/SSL where required, backup storage, and monitoring.
- **Knowledge resources:** project requirements, exercise catalogue, UX standards, API documentation, database documentation, and testing standards.

8. Risk Management Plan

Risk	Impact	Mitigation
Requirement changes	Medium/High	Freeze baseline requirements after review; track change requests
Schedule delay	High	Prioritize MVP features and monitor milestones weekly
Data/privacy issue	High	Role-based access, validation, encryption in transit, secure storage
AI feedback quality	Medium/High	Use constrained prompts, validation, human review, and clear limitations
Integration problems	Medium	Define APIs early and perform integration testing continuously
Performance issues	Medium	Use efficient queries, pagination, caching where appropriate, and load testing
Deployment failure	Medium	Maintain staging environment, backups, rollback procedure, and deployment checklist
Scope creep	High	Use formal scope control and prioritize features by project value

9. Quality Management Plan

- Define acceptance criteria before development begins for each major feature.
- Use code review and version control for development changes.
- Run automated or repeatable tests for critical business logic.
- Test responsive behavior across common screen sizes.
- Maintain a defect log with severity, owner, status, and resolution.
- Conduct regression testing after major changes.
- Use user acceptance testing to confirm athlete and coach workflows.

10. Communication and Monitoring Plan

Activity	Frequency	Purpose
Project status review	Weekly	Review progress, blockers, risks, and upcoming work
Development sync	2–3 times/week	Coordinate technical tasks and dependencies
Design review	At key milestones	Validate UI/UX against requirements
Testing review	During testing phase	Review defects and release readiness
Stakeholder/demo review	End of major phase	Collect feedback and approve milestones

11. Change Management

- Record each requested change with description, reason, affected modules, and priority.
- Estimate the effect on schedule, resources, testing, and existing functionality.
- Approve significant changes before development work begins.
- Update requirements, design, test cases, and documentation when a change is accepted.
- Use version control so previous working versions can be recovered when necessary.

12. MVP Prioritization

Priority	Features
Must Have	Registration/login, roles, athlete profile, exercise catalogue, workout plans, workout sessions, basic progress dashboard
Should Have	Coach dashboard, feedback workflow, plan revision history, filtering/search
Could Have	Notifications, exercise videos, advanced analytics, wearable integration
Future	Mobile app, advanced recommendation models, broader integrations

13. Project Success Criteria

- All approved MVP requirements are implemented and documented.
- Athletes can register, view a plan, complete a workout, and review progress.
- Coaches/administrators can manage athletes and workout plans according to their permissions.
- Critical workflows pass functional and integration testing.
- Security and access controls are verified.
- The application is deployable using documented setup steps.
- The final system is understandable, maintainable, and ready for future enhancement.

14. Final Project Planning Summary

The FitBuddy project should be executed in controlled phases: define requirements, design the user experience and architecture, build the MVP, integrate the feedback system, test the complete workflow, deploy, and document the result. Regular milestone reviews, risk tracking, quality checks, and scope control will help keep development organized while leaving room for future capabilities such as wearable integration, notifications, and advanced analytics.